

SACRAMENTO EXECUTIVE, CA, USA

WMO: 724830

Lat: 38.507N

Lon: 121.495W

Elev: 15

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23232

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.5	34.0	20.3	15.3	53.2	24.1	18.3	51.3	24.8	54.3	21.2	52.0	1.1	330	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.8	100.2	69.6	97.0	68.3	93.7	67.3	72.1	95.1	70.3	92.5	68.7	90.0	6.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	87.9	83.1	61.5	81.7	79.1	60.1	77.6	76.8	35.7	94.9	34.1	92.8	32.8	90.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	17.5	15.3	27.6	106.1	2.8	2.5	25.6	107.9	24.0	109.3	22.4	110.7	20.4	112.5
			26.4	74.3	2.8	2.3	24.4	75.9	22.7	77.3	21.2	78.6	19.1	80.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	47.9	51.3	55.5	59.3	65.7	72.0	75.7	75.0	72.1	64.0	53.9	47.6
	DBStd	11.38	4.65	4.54	5.19	5.99	6.17	6.03	5.16	4.75	5.11	5.25	5.36	5.06
	HDD50	278	95	36	12	3	0	0	0	0	0	0	23	109
	HDD65	2436	530	385	297	190	64	9	0	0	5	81	334	541
	CDD50	4556	30	71	181	282	488	660	796	777	663	435	140	33
	CDD65	1238	0	0	1	20	86	219	331	312	218	50	1	0
	CDH74	15445	0	1	61	364	1276	2836	3973	3611	2550	745	28	0
	CDH80	7894	0	0	3	107	552	1474	2238	1985	1291	243	1	0
(14) Wind	WSAvg	6.0	4.9	5.7	6.2	6.9	7.3	7.4	7.1	6.5	5.5	5.1	4.3	5.1
(15) Precipitation	PrecAvg	19.8	3.7	3.5	2.9	1.4	0.8	0.2	0.0	0.0	0.1	1.1	2.0	4.0
	PrecMax	32.8	9.5	10.9	8.7	4.4	3.7	1.5	0.2	0.5	0.5	3.9	4.6	9.8
	PrecMin	7.8	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.2
	PrecStd	5.5	2.9	2.4	2.5	1.0	0.9	0.4	0.0	0.1	0.2	0.9	1.2	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	71.9	79.6	88.5	96.6	103.0	104.2	104.1	100.2	92.3	77.7	64.6
		MCWB	52.3	57.3	61.3	63.6	68.1	70.8	71.6	70.2	67.6	64.7	59.6	54.8
	2%	DB	61.8	67.4	75.2	82.9	91.2	97.9	99.9	98.9	96.0	86.1	72.9	61.2
		MCWB	53.0	55.9	59.4	61.1	66.0	68.9	70.3	69.2	66.5	62.4	57.9	54.0
	5%	DB	58.9	63.8	71.4	78.2	86.5	93.6	96.1	95.1	92.2	81.9	68.1	58.9
		MCWB	52.5	53.9	57.7	60.3	63.2	67.0	68.8	67.8	65.3	60.3	56.2	52.8
	10%	DB	56.2	60.6	67.4	73.0	81.4	88.6	92.2	91.0	88.0	77.7	64.2	56.6
		MCWB	51.2	52.9	55.9	57.9	61.7	65.8	68.1	66.9	64.4	59.2	54.7	51.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	60.5	63.9	67.2	70.6	73.3	75.3	75.2	71.6	67.3	62.5	59.1
		MCDB	60.9	66.4	75.9	81.5	91.3	97.6	98.4	97.1	92.6	85.2	73.1	62.3
	2%	WB	55.6	58.2	61.3	63.8	67.7	70.8	72.1	71.1	68.5	64.6	60.0	56.8
		MCDB	58.7	64.3	72.5	78.8	88.0	93.7	95.6	94.2	89.1	81.6	68.6	59.3
	5%	WB	53.6	56.1	59.2	61.6	65.3	68.7	70.3	69.1	66.9	62.6	58.2	54.6
		MCDB	56.4	61.4	68.8	75.4	83.6	90.0	92.7	91.2	87.6	77.3	65.2	57.0
	10%	WB	52.1	54.1	57.0	59.4	63.0	66.8	68.6	67.6	65.4	60.8	56.3	52.5
		MCDB	55.2	58.9	65.4	70.9	78.6	86.4	89.9	88.7	85.6	73.7	62.3	55.3
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	17.9	20.7	23.2	26.4	29.7	31.8	31.5	30.2	26.3	20.3	16.0
		MCDBR	20.6	24.0	27.9	31.7	34.5	36.6	36.4	36.7	36.6	33.4	26.9	19.3
	5% WB	MCWBR	12.6	14.0	14.5	14.1	13.2	13.0	13.1	12.6	12.0	12.5	13.4	11.9
		MCWBR	13.8	19.3	24.9	28.7	32.6	34.2	34.5	33.8	32.6	28.6	21.7	11.5
(40) Clear-Sky Solar Irradiance	taub	0.324	0.326	0.341	0.356	0.365	0.362	0.363	0.347	0.351	0.345	0.331	0.333	0.333
		taud	2.503	2.504	2.449	2.399	2.395	2.413	2.407	2.442	2.437	2.463	2.486	2.473
	Ebn at Noon	271	286	292	292	290	290	289	292	284	275	266	258	258
		Edh at Noon	25	28	33	37	37	37	37	35	33	29	25	24
(44) All-Sky Solar Radiation	RadAvg	685	1025	1464	1939	2352	2618	2603	2332	1892	1338	857	616	616
	RadStd	130	128	148	161	138	102	73	73	64	85	60	102	102

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	-0.94	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	+0.72	N/A	-1.94	N/A	N/A	N/A	N/A	-209	+248	+76	

Nomenclature: See separate page